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Model-based parameter optimization for arc welding process simulation ☆

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ABSTRACT

This paper presents a model-based parameter optimization for simulating a metal-inert gas welding process. The computational model used in this study is based on computational fluid dynamics methods and implemented using the finite volume approach on a 3D computational domain. The wire electrode, the arc plasma and the workpiece are treated as a self-consistent system. Important welding parameters, including arc current, wire feed rate, workpiece thickness, welding speed and geometry, as well as the metal alloy types used for the wire and workpiece, were implemented as adjustable parameters. By tuning these parameters, the performance of the arc welding can be predicted, and different settings can be compared to optimize welding performance.

A benchmarking study of the arc model against experimental measurements is presented to demonstrate the model's capabilities in the prediction of the weld pool changes and thermal dynamics involved in the welding process. Two numerical case studies are presented to demonstrate the use of the model-based optimization to quantify welding pool variations with the change in welding parameters. The first case study is the determination of the optimal arc current and welding speed settings for different workpiece thicknesses. The optimization process shows that the predictions are not only in agreement with established experimental welding experience on the direct relationship between workpiece thickness and arc current, but more importantly quantify this relationship for a given workpiece thickness. The second case study focuses on the welding parameters optimization for different metal alloys. The comparison suggests that the welding parameters suitable for some aluminium alloys are less likely to be successful in welding magnesium alloys. A further model validation of Mg alloy AZ31 welding shows an agreement with experimental measurements. The work presented shows the potential of model-based parameter optimization to assist process engineers in the practical improvement of the welding process.

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1. Introduction

Metal inert-gas (MIG) welding is a common arc welding process used to join metal pieces together using the high temperature and heat flux generated by an electric arc between a wire anode and a workpiece cathode. It is used in a wide range of engineering sectors, including manufacturing, rail, automotive, energy and construction [1]. MIG welding involves many process parameters, such as arc current, workpiece thickness and welding geometry, wire electrode feed rate, type of shielding gas, as well as the alloys selected for the wire electrode and workpiece. Any of those parameters can influence the final quality of the welded products. As a commonly used technology, there are comprehensive guidelines and industrial protocols for appropriately setting up welding equipment [2] and effectively selecting welding parameters in different applications [3,4]. Welding parameter optimization is crucial for improving welding quality and increasing productivity [5]. However, the optimization process in practice is heavily dominated by welding trials, relying on empirical knowledge and experience. With the rapid innovation in new metals and materials and the increasing demanding for customized design and applications, the welding industry needs more effective means for process optimization directed towards producing new material designs and customized applications.

Modelling approaches have become an attractive option for welding parameter optimization. A modelling tool can help a user set up virtual welding experiments and predict appropriate parameter settings prior to physical tests. This is ideal for situations where experimental knowledge is limited, for example, if a new type of metal is introduced or an uncommon welding geometry is applied. Such virtual testing can provide the estimated parameter ranges for guiding the experimental setup. Moreover, the computational models can effectively take into account empirical knowledge as model inputs to improve the numerical predictions. Such feedback interactions between the modelling approach and experimental knowledge can greatly improve welding effectiveness and productivity.

Many computational models of arc welding have been presented in the literature. Most of the models have limitations, such as including only the workpiece in the computational domain [6,7], assuming a flat weld pool surface [8], models of droplet formation [9] and transfer into the weld pool [10–12] that are restricted to two dimensions, or three-dimensional models of arcs that exclude the weld pool [13,14]. In the past several years, CSIRO has developed a three-dimensional computational model of MIG welding in a computational domain that includes the wire electrode, arc and workpiece self-consistently [15–17]. The MIG model uses computational fluid dynamics techniques to model the interaction between the arc plasma and the workpiece by solving conservation equations of mass, energy and momentum. The formation of droplets and their transfer into the weld pool has been modelled and the deformation of the welding pool surface due to droplet transfer has been calculated. The effects of metal vapour produced from the wire electrode and workpiece have also been taken into account [16]. Mixing of the wire and workpiece alloys in the weld pool has also been included, and recently the mushy zone has been considered [18]. To promote the existing arc welding models for general use by welding researchers and engineers, the MIG welding model has been streamlined into a software application that can be easily launched from a desktop computer [19,20]. Important welding parameters have been designed as user-adjustable, and the corresponding graphical interfaces have been provided for taking user inputs. The software package, named *ArcWeld*, has been initially applied in commercial projects in the automotive industry and recently extended for application in the rail industry [18].

This work presents a benchmarking study using the *ArcWeld* software in order to quantify the impact of welding processing parameters on the predicted performance. The objective of the study is to gain a better understanding of the arc welding process through theoretical modelling and numerical analysis, and ultimately provide critical information and practical guidelines for welding process optimization. Experimental benchmarking will be presented to validate the model-predicted welding pool depth with the measured cross-section depth. Two case studies were conducted to demonstrate the model-based optimization process in order to study the interrelationships of arc current, workpiece thickness, welding speed, wire feed rate, and the alloys selected for the wire electrode and workpiece. The first case study will be presented to find the optimal arc current settings for different thicknesses of workpiece. The optimization process shows that the predictions are not only in agreement with established experimental welding experience on the direct relationship between workpiece thickness and arc current, but more importantly quantify this relationship for a given workpiece thickness and desired weld pool depth. The second case study will focus on the welding parameters optimization for different metal alloys. It shows that the model-based optimisation could be useful to guide the experimental settings for uncommonly used metals or new materials. The work presented shows an effective way to identify key parameters in the complicated welding process and quantify their effects on the weld quality.

The technical details of the existing MIG welding models will be briefly described in the next section, and the *ArcWeld* software will be introduced and demonstrated in Section 3. The experimental benchmarking and case studies on parameter optimization will be presented in Section 4, followed by a conclusion in Section 5.

2. MIG welding modelling approach

The MIG welding model uses computational fluid dynamics techniques to solve heat transfer equations, energy conservation equations, mass continuity and momentum conservation equations in a three-dimensional computational domain [15]. A schematic view of two workpieces welded in a flat bead-on-plate (BOP) geometry is shown in Fig. 1(a). The computational domain consists of the wire electrode, arc plasma and workpiece and is represented by 3D mesh points in a Cartesian coordinate system, as shown in Fig. 1(b).

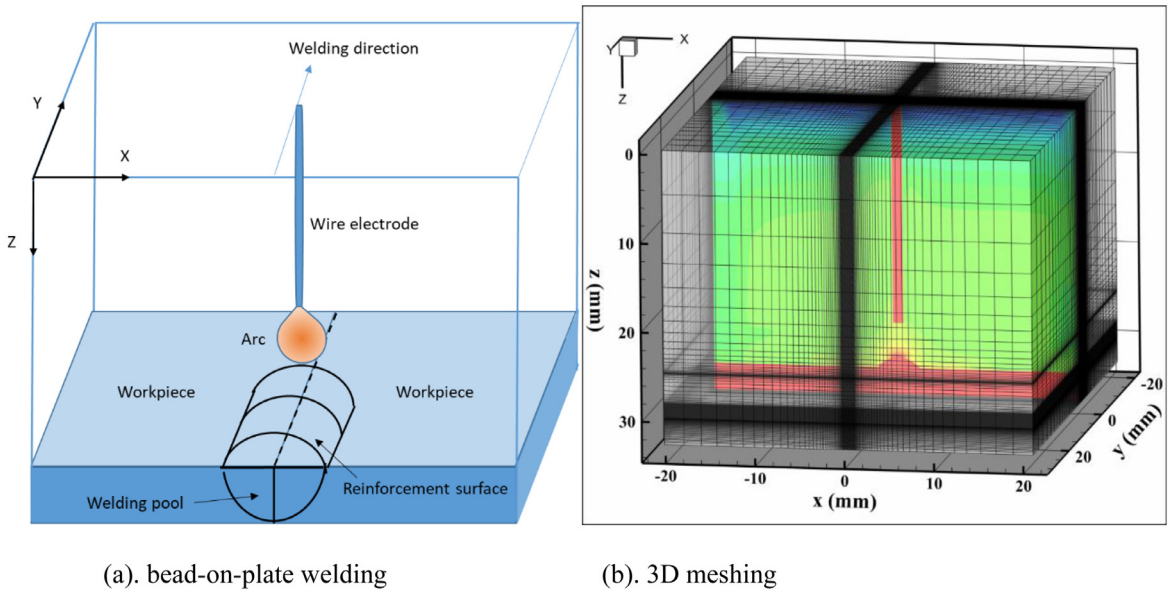


Fig. 1. Schematic view of a bead-on-plate welding (a) and mesh representation of the computational domain (b).

The mass continuity equation is given by Murphy et al. [17]:

$$\nabla \cdot (\rho \boldsymbol{v}) = S_M \tag{1}$$

where ρ is mass density and $\boldsymbol{v}$ is the fluid velocity. The source term S_M here describes the production of metal vapour from the wire electrode, workpiece and droplets. The equation of momentum conservation is defined as [17]:

$$\nabla \cdot [\rho(\boldsymbol{v} - \boldsymbol{V})(\boldsymbol{v} - \boldsymbol{V})] = -\nabla P - \nabla \cdot \boldsymbol{\tau} + \boldsymbol{j} \times \boldsymbol{B} + \rho \boldsymbol{g} \tag{2}$$

where P is the pressure, $\boldsymbol{\tau}$ is stress tensor and $\boldsymbol{g}$ is the acceleration of gravity. The cross product of current density $\boldsymbol{j}$ and magnetic field $\boldsymbol{B}$ defines the Lorentz force exerted by the electromagnetic field. The energy conservation equation is [17]:

$$\nabla \cdot [\rho(\boldsymbol{v} - \boldsymbol{V})h] = \frac{\boldsymbol{j}^2}{\sigma} - \nabla \cdot (\alpha \ \nabla h) - \nabla \cdot [(h_m - h_s) \ \alpha \ \nabla Y_M] - \frac{5k_B}{2ec_p}\boldsymbol{j} \cdot \nabla h - U - S_v \tag{3}$$

where h, h_m, and h_s denote specific enthalpy of the modelled system, of metal vapour and of shielding gas respectively. The factor $\alpha = {}^{k}/_{c_p}$ is a thermal diffusion related coefficient, in which thermal conductivity k is divided by specific heat c_p at constant pressure. The effect of metal vapour on thermal transport is considered; Y_M is the metal vapour mass fraction. The fourth term on the right-hand side describes electron diffusion; k_B and e are Boltzmann's constant and electronic charge. The last two terms on the right-hand side are the net radiative emission coefficient U and the latent heat of vaporization source term S_v. The last four terms are only considered in the arc plasma.

In Eqs. (2) and (3), $\boldsymbol{V}$ is the welding velocity, i.e. the velocity of the wire electrode with respect to the stationary workpiece. Arc welding is a steady-state process if the welding velocity, the wire feed rate and arc current are constant and if the influence of the periodic droplets is averaged over time; this is clear from the fact that the weld properties are uniform along the weld seam. Based on the work of Mundra et al. [23], our model uses a coordinate system that moves with the heat source. Under this coordinate system, the heat source is stationary in space, with the metal entering and leaving the computational domain at the welding speed $\boldsymbol{V}$. In this frame of reference, the governing equations are solved for net velocity, i.e. combined effect of convective velocity and the welding velocity. In our model, the welding velocity, the wire feed rate and arc current are constant, so the properties of the arc and weld pool do not change with time, making a steady-state model possible and reducing the computational complexity.

Two further conservation equations are solved, for the mass fraction of metal vapour in the arc plasma and for the mass fraction of the wire (or droplet) alloy in the weld pool [17]:

$$\nabla \cdot (\rho \boldsymbol{v} Y_M) = -\nabla \cdot \boldsymbol{J}_M + S_M \tag{4}$$

$$\nabla \cdot (\rho \boldsymbol{v} Y_d) = -\nabla \cdot \left(\rho D_d^l \nabla Y_d\right) + S_d \tag{5}$$

where $\boldsymbol{J}_M$ is diffusion mass flux of metal vapour, D_d^l is self-diffusion coefficient of liquid metal. Y_d and S_d are mass fraction of droplet alloy and droplet alloy source term, respectively.

Finally, equations for current continuity and to calculate the magnetic field [17] have to be solved:

$$\nabla \cdot \boldsymbol{j} = \nabla \cdot (\sigma \nabla \emptyset) = 0 \tag{6}$$

$$\nabla^2 \boldsymbol{A} = -\mu_0 \boldsymbol{j} \tag{7}$$

where $\emptyset$ and $\boldsymbol{A}$ are electric potential and magnetic potential, σ is electrical conductivity and μ_0 is permittivity of free space.

The influence of droplets is calculated using a method that averages the two-way interactions between the droplets and the arc and weld pool over time and over the cross-sectional area of the droplets [21]. For the highest welding speed (0.9 m/min) and the standard droplet frequency (93 Hz) used in this study, the wire and arc move 0.16 mm between droplets, which is much less than the typical droplet diameter of 1 mm. It is therefore reasonable to average the influence of the droplets in time. The shape of the boundary between the weld pool and the arc plasma is calculated by minimizing the energy of the surface, taking into account surface tension, the pressure exerted by the arc and the droplets, buoyancy and the volume of metal transferred by the droplets [15].

A recent improvement introduced consideration of the mushy zones into the model, which give more accurate prediction of the temperature variations within the mixed solid-liquid regions (i.e. the mushy state) in the weld pool [18]. Consideration of the mixing of the droplets from the wire and the workpiece alloy in the weld pool means to allow the temperature field in the mushy zone to be calculated more accurately, taking into account the dependence of the solidus and liquidus temperatures on the alloy composition. This greatly improves the accuracy of the predicted size and shape of the welding pool.

The finite volume method (FVM) [22] has been used to solve the equations in a 3D mesh. Variable meshing techniques have been applied to the computational domain (Fig. 1(b)), in which finer meshing is created for regions where the arc plasma interacts with the wire electrode (denser volume elements around the wire electrode) and the workpiece (denser volume elements around the workpiece surface area), while coarser meshing is applied elsewhere. This gives a good balance between preserving boundary details and reducing computational cost.

Boundary conditions for both the external and internal boundaries are applied in the FVM solver. The external boundaries are the six planes bounding the computational domain, as shown in Fig. 1(a). At the bottom of the workpiece, the initial values for the electrical potential, the gradient of the magnetic potential and the flow velocity are set to zero. The heat flux coefficient is set to 50 W m^{-2} K^{-1}. At the other external boundaries, the following boundary conditions are used. The gradient of the electric potential is set to zero, except for the top of the electrode, where it is defined as $I/\pi\sigma r^2$ (arc current I, electrical conductivity σ and radius r of the electrode). The gradient of the magnetic potential is set to zero. The flow velocity is set to zero, except for the top region surrounding the electrode, where the input gas flow comes through a nozzle. The boundary temperature is set to 300 K.

Two internal boundaries are considered: one between the arc plasma and the wire electrode and another between the arc plasma and the workpiece. Heat transfer between the arc plasma and the electrode boundary is calculated using the LTE-diffusion approach of Lowke and Tanaka [24]. To deal with the departures from LTE that occur in the plasma boundary [25,26], a wide range of treatments of the region has been investigated [27,28]. These include a two-dimensional two-temperature and chemical non-equilibrium model to investigate arc–anode attachment [29] and a two-dimensional time-dependent model taking into account deviations from thermal equilibrium [30,31]. However, Lowke and Tanaka's numerical study [24] showed that the LTE-diffusion approach allows the arc voltage and the heat transfer to the electrode to be calculated accurately by choosing a mesh size that is equal to the electron diffusion at the arc–electrode boundary. In particular, the LTE-diffusion model uses a mesh size near the electrode boundary that is set to a value D_e/W, called the electron diffusion length [24], where D_e is the electron diffusion coefficient and W is the electron drift velocity. Lowke and Tanaka [24] pointed out that at the electrode surface, the electron density gradients are extremely large and the diffusion term will be dominant. They showed that using this diffusion length as the mesh size at the electrode surface avoids having to resolve the region dominated by diffusion effects. Their approach provided results in agreement with experiment and more sophisticated models that accounted for deviations from LTE. We use this LTE-diffusion approach in our *ArcWeld* model to ensure acceptable accuracy and fast computation. The latter is important for practical application of the software in an industrial environment.

For points on the wire electrode boundary adjacent to the plasma, an additional term is added to the source term of Eq. (3) [15]:

$$S_e = j\emptyset_{we} - \varepsilon_e k_B T_e^4 - S_v \tag{8}$$

The first term in the right side denotes the heating effect of electrons absorbed at the electrode, the second term defines radiative cooling of the electrode and the third team is the cooling due to vaporization of the liquid metal ($\emptyset_{we}$ is the work function of the electrode material; ε_e and T_e are the emissivity and the surface temperature of the electrode; k_B is the Stefan–Boltzmann constant). Details of the calculation of vaporization term, and the energy that is transferred to the plasma due to the flux of metal vapour, were given previously [16].

Similarly, for points on the plasma and workpiece boundary, an additional term is also added to the energy conservation Eq. (3) [15]:

$$S_w = j_w(V_w - \emptyset_{ww}) - \varepsilon_w k_B T_w^4 - S_v \tag{9}$$

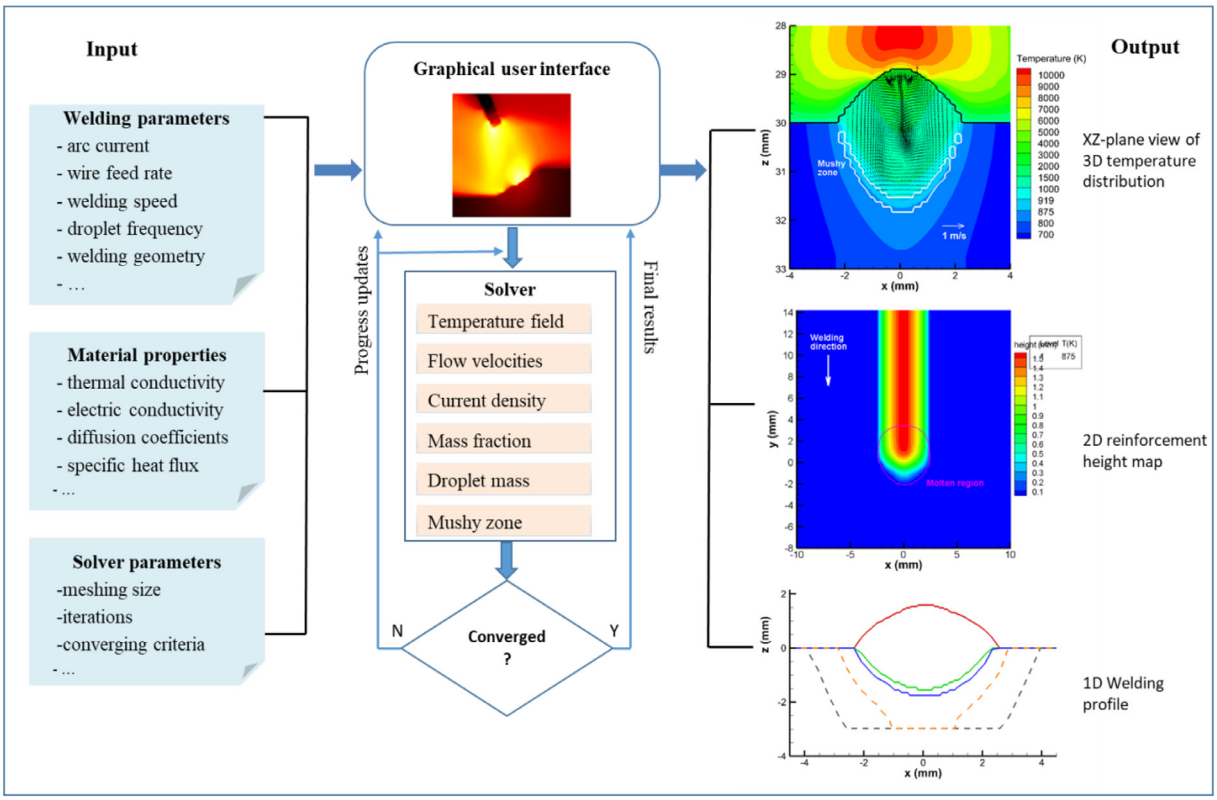


Fig. 2. Data flow of the MIG welding modelling software and the graphical presentation of the simulation results.

Here the first term represents the heating effect of the species accelerated in the sheath voltage less the energy required to emit electrons from the workpiece, while the second term is the blackbody radiation loss and is generally negligible and the third term is the latent heat of vaporization (j_w is the local current density; V_w is the cathode fall voltage and $\emptyset_{ww}$ is the work function of the workpiece material; ε_w and T_w are the emissivity and the surface temperature of the workpiece).

The properties calculated by the finite volume solver of the MIG welding models include temperature, current density, flow velocity, electric and magnetic potential, metal vapour mass fraction and droplet mass fraction. Using the modelling approach, interactions among these properties and their interrelationships can be derived or estimated, which is of crucial importance to the understanding of arc plasma physics in MIG welding, yet difficult and in some cases impossible to be measured experimentally.

3. Application software *ArcWeld*

MIG welding models have been streamlined as an application software, named *ArcWeld*, for research and industrial use [19]. The *ArcWeld* software was built as a flexible graphical user interface (GUI) in front end and a computing-intensive solver back end (illustrated in Fig. 2). The input data includes welding parameters, material property data and solver-related parameters. Of these, the welding and solver parameters are user adjustable and directly linked with the GUI to obtain user inputs, while the properties of the user-selected metal alloys are derived from either theoretical calculations or empirical observations in the literature and fed into the solver modules as the initial conditions. The boundary conditions are dependent on the welding geometry, which can be selected from a pre-configured geometry list, including flat bead-on-plate, lap-fillet, T-joint and corner-joint geometries. The wire electrode work and travel angles can also be selected by the user. The *ArcWeld* software application GUI was designed to provide a central hub for parameter control, solver execution, result display and progress monitoring.

Once the workpiece thickness and welding geometry are selected, the computational domain of the solver is defined and subdivided into a 3D mesh for finite volume analysis [22]. Internal boundary conditions are applied in regions where the plasma interacts with the wire tips, as well as with the workpiece surface. Finer meshing is applied to the boundary regions to improve accuracy in predicting boundary transitions, as shown in Fig. 1(b). The solver starts from initial conditions and boundary constraints, then solves iteratively on 3D grids until a converged solution is obtained. Thermophysical properties and weld pool contours are re-calculated during iterations to take into account the evolution of thermal dynamics and pool formation in the MIG welding process. The shape of the weld pool surface (the boundary between the weld pool and the

plasma, i.e. the reinforcement height) is calculated regularly to check if it has reached a converged state (i.e. the change in the reinforcement height is less than a given limit). The solver is terminated when convergence is reached and the final simulation results are displayed in the GUI.

Three types of output data are produced by the *ArcWeld* software (shown from top to bottom on the output side of Fig. 2): 2D slices of 3D data showing thermal and physical properties such as temperature and velocity; a 2D colour-coded map showing the reinforcement height in welded regions; and the welding cross-section showing the reinforcement height, the liquidus and solidus depths and user-defined heat-affected zones. The mushy zone is defined as the region between the solid and liquidus curves, as highlighted in the temperature distribution.

4. Benchmarking

4.1. Experimental data validation

To validate the welding modelling software, we have conducted benchmarking of the model-predicted welding profiles against weld cross-sectional measurements provided by our previous collaborators from the automotive industry. Similar comparisons have been published previously [17,20]. However, recent improvements to the *ArcWeld* solver have been made, including the consideration of the mushy zone and an update of the alloy thermophysical properties, so it is important to perform the benchmarking validation again.

An ABB Robot was used in the welding experiments to control the movement of welding torch. The distance between the torch to the workpiece is 12 mm. The shielding gas is 100% argon and the welding mode is pulsed transfer. The welding experiments were conducted for both bead-on-plate geometry with a vertical wire electrode and lap-fillet geometry with 60° work angle and 90° travel angle relative to the horizontal.

The main parameters used for experiments and the numerical simulation are given in Table 1. The experimental results have been measured from the weld cross-section at three positions – position 1 (25 mm), position 2 (114 mm) and position 3 (203 mm) from the start of the weld. Fig. 3 shows the cross-sectional macrographs of a bead-on-plate (left) weld and a

Table 1
The main welding parameters used for both experimental measurements and numerical simulations. Alloys used for wire electrode and workpiece cathode are aluminium–silicon alloy AA4043 (Al + 5.3 wt% Si) and aluminium–magnesium alloy AA5754 (Al + 3.1 wt% Mg) [32].

Parameter	BOP geometry	Lap-fillet geometry
Arc current (A)	95	104
Gas flow (L/min)	14.16	14.16
Wire feed rate (m/min)	4.3	4.7
Droplet frequency (1/s)	93	93
Welding speed (m/min)	0.9	0.6
Workpiece thickness (mm)	3	3
Wire anode alloy	AA4043	AA4043
Workpiece cathode alloy	AA5754	AA5754
Wire electrode angle	90° (vertical)	Work angle 60°, travel angle 90°

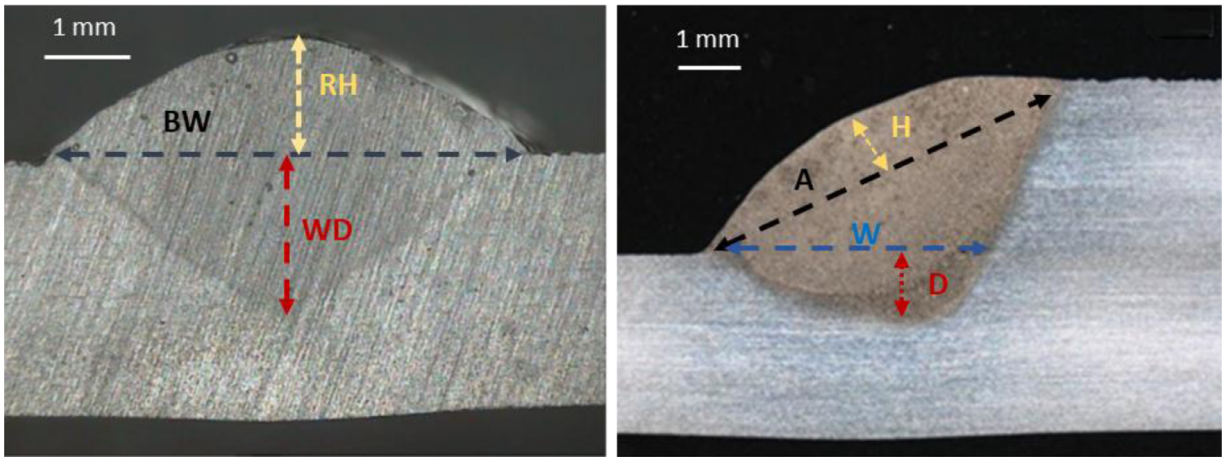


Fig. 3. Cross-sectional graphs of the welded parts cut at 25 mm position from the start of the welds for bead-on-plate geometry (left) and lap-fillet geometry (right). The arrows marked on the images represent the measured weld dimensions.

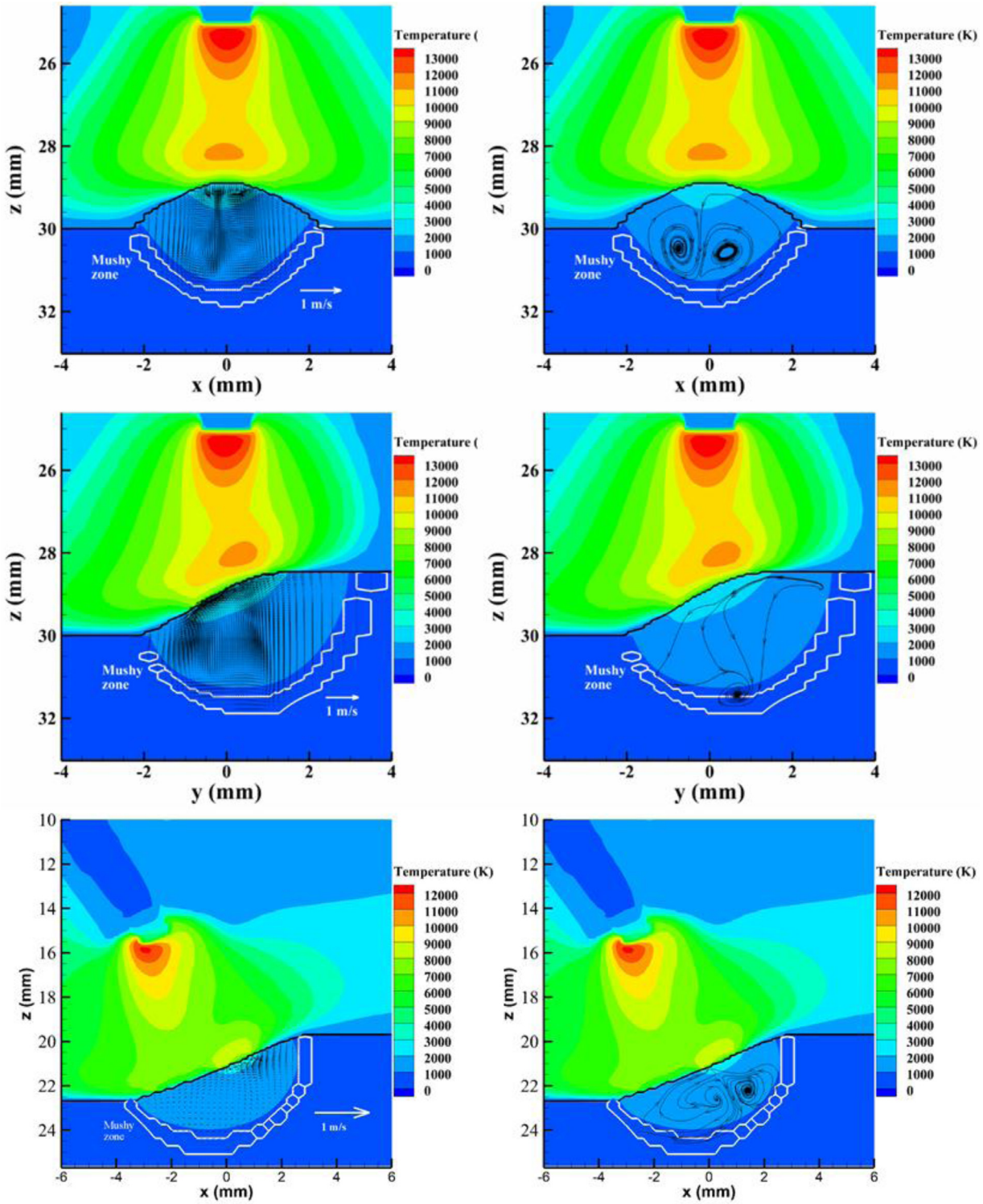


Fig. 4. Predicted temperature distribution and the flow velocity in weld pool. Top row: temperature and flow fields in cross-sections in the x–z plane for bead-on-plate geometry. Middle row: temperature and flow fields in cross-section in the y–z plane for bead-on-plate geometry. Last row: temperature and flow fields in cross-section in the x–z plane for lap-fillet geometry. Vectors are used in the figures on the left to show magnitude and distribution of the flow, while on the right, streamlines are used to show the direction of the flow more clearly.

Table 2
Numerical comparison of the measured weld dimensions with the model predicted values for both bead-on-plate and lap-fillet geometries.

		RH (mm)	WD (mm)	BW (mm)	WD/T	RH/BW
BOP	Measured	1.47	1.85	5.92	0.62	0.25
	Model-predicted	1.54	1.76	5.54	0.59	0.28
		A (mm)	**H (mm)**	**W (mm)**	**D (mm)**	
Fillet	Measured	7.12	1.26	5.07	1.42	
	Model-predicted	8.15	1.42	6.37	1.67	

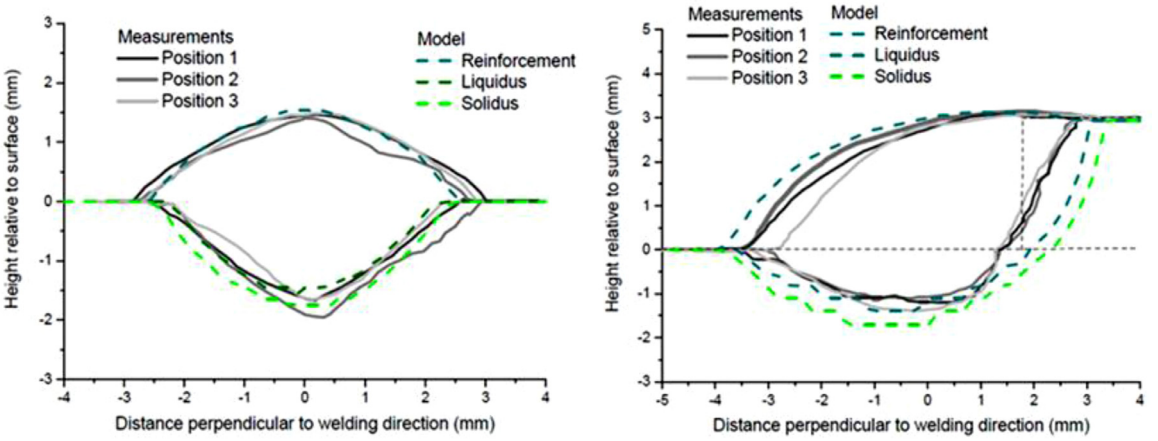


Fig. 5. Comparison of the model-predicted reinforcement height and weld depths of the cross-section measurements for both bead-on-plate geometry (left) and lap-fillet geometry (right). Positions 1, 2 and 3 in measurements are respectively 25, 114 and 203 mm from the start of the weld.

lap-fillet (right) weld at position 1. The weld pool shape can be clearly observed from the colour difference between the resolidified region and the remaining workpiece. The reinforcement height (RH), weld depth (WD) and the bead width (BW) for the bead-on-plate weld were measured, as shown by the arrows marked on the left image, while the measurements for the lap-fillet weld are marked on the right image.

The measured weld dimensions and the predicted values are given in Table 2 for both bead-on-plate geometry and lap-fillet geometry. Values for the experiments are the average of three measurements at different positions along a weld. The simulation results for both bead-on-plate and lap-fillet geometries are shown in Fig. 4. These cross-section views show the temperature distribution in the arc and the flow velocity in the weld pool. The first two rows in the figure show the cross-sections in the x–z and y–z planes of the bead-on-plate weld. The last row shows the cross-section in the x–z plane of the lap-fillet weld. The flow velocity in the weld pool region is represented in both a vector (left figures) and a streamline (right figures) plot to show the magnitude and the direction of the flow.

The flow velocity vectors indicate a downward flow at the top of the weld pool due to the momentum transferred from the wire droplets to the weld pool. The mushy zone boundaries are marked in white at the bottom of the weld pool to indicate that the temperature within these regions is between the liquidus and the solidus temperatures. As expected, the flow velocities are low in the mushy zone because of the frictional dissipation of the flow.

Fig. 5 compares three cross-sectional measurements with the model-predicted weld profiles. The solid lines represent the measurements of the reinforcement heights (top curves) and the weld pool contours (bottom curves) at three locations. The dashed lines represent the model predicted reinforcement height at the top and the liquidus and the solidus contours at the bottom. As can be seen from Table 2 and Fig. 5, the reinforcement height and the weld pool depth are predicted accurately by the *ArcWeld* model for BOP geometry and slightly over-predicted for the lap-fillet geometry.

It is important to note that the experimental measurements can vary from weld to weld significantly. Even in the same weld, as shown in Fig. 5, notable variability in the cross-sectional measurements is observed at different positions for both geometries. Larger variability, sometimes greater than 1 mm, can occur between different welds.

4.2. Finding optimal arc current settings for different workpiece thicknesses

Arc current is a very important parameter in MIG welding. The electric arc between the consumable wire electrode and metal workpiece creates a circuit that produces the heat required to melt and join metals. The value of the arc current affects the magnitude of the current density, the magnetic field induced by the current and the temperature field. As a rule of thumb, higher arc current is usually used for welding thicker workpieces to generate higher temeprtaure field. In such

Table 3
The main welding parameters used for arc current optimisation for different workpiece thicknesses in BOP geometry.

Parameter	Value
Arc current (A)	(a) 95, (b) 150, (c) 250
Wire feed rate (m/min)	(a) 4.3, (b) 6, (c) 10.4
Welding speed (m/min)	(a) 0.9, (b) 0.9, (c) 0.7
Gas flow (L/min)	14.16
Droplet frequency (1/s)	93
Workpiece thickness (mm)	3, 5 and 8
Wire diameter (mm)	1.2
Wire anode alloy	AA4043 (Al + 5.3 wt% Si)
Workpiece cathode alloy	AA5754 (Al + 3.1 wt% Mg)
Wire electrode angle	90° (vertical)
Welding geometry	Bead-on-plate

Table 4
Model predicted results for different work piece thicknesses simulated at different arc current settings. The optimal settings are highlighted in bold for each workpiece thickness.

Thickness (mm)	Arc current (A)	Wire feed rate (m/min)	Welding speed (m/min)	RH (mm)	WD (mm)	$\gamma = WD/T$
3	**95**	**4.3**	**0.9**	**1.54**	**1.76**	**0.59**
	150	6	0.9	1.21	3	1
	250	10.4	0.7	–	–	–
5	95	4.3	0.9	1.65	1.19	0.24
	150	**6**	**0.9**	**1.48**	**2.96**	**0.59**
	250	10.4	0.7	2.1	5	1
8	95	4.3	0.9	1.65	1.04	0.13
	150	6	0.9	1.71	2.2	0.28
	250	**10.4**	**0.7**	**2.09**	**4.65**	**0.58**

a case, other welding parameters can also be adjusted, such as reducing the welding speed, in order to generate a deeper welding pool.

In this section, the model-based optimization process will be demonstrated by using the model to quantify the relationship between the arc current and the workpiece thickness for bead-on-plate welding geometry. The optimization process usually requires the selection of an objective function or indicator so that the model parameters can be optimized towards achieving this objective. MIG welding is an application-dependent technique, which implies that the objective function will depend strongly on the technical requirements of a specific application. Based on the experimental validation case presented in Section 4.1, a ratio of the welding depth and the workpiece thickness, $\gamma = WD/T$, of 0.6 gives a good bead-on-plate weld. In this case study, we will use γ as the objective indicator and extend the required range to between 0.55 and 0.75 (i.e. from just over one-half to three-quarter penetration) to allow for the experimental variations that occur in practice. It should be noted that in some applications, full penetration can be a desirable result, in which case the target range for γ would be set to close to 1.

To demonstrate the model-based optimization process for finding the optimal arc current settings for different workpiece thicknesses, numerical simulations were conducted for three different workpiece thicknesses – 3 mm, 5 mm and 8 mm. For each workpiece thickness, three arc current settings were applied, referred to as the low arc current setting (setting (a): arc current 95 A, wire feed rate 4.3 m/min, welding speed 0.9 m/min), middle arc current setting (setting (b): arc current 150 A, wire feed rate 6 m/min, welding speed 0.9 m/min) and high arc current setting (setting (c): arc current 250 A, wire feed rate 10.4 m/min, welding speed 0.7 m/min). Table 3 lists the three arc current settings and other common parameters used in the simulations. Table 4 shows the simulation results of each thickness under low, middle and high arc current settings. The wire feed rate has an approximately linear relationship to arc current in order to maintain a constant arc length. It is derived based from data provided by Lincoln Electric [34]. The welding speed is chosen to be relatively low for 250 A since such a high arc current is mainly used for thicker workpieces, which need longer penetration time.

The ratio γ of the welding depth to the workpiece thickness shown in Table 4 indicates that for the 3 mm workpiece, the low arc current setting produces a good weld (as validated in Section 4.1) with $\gamma = 0.59$, inside the target range of 0.55–0.75, while the middle current settings are too high, giving penetration through the bottom of the workpiece ($\gamma = 1$). In this case high arc current settings did not require testing since penetration will be likewise too high.

For the 5 mm workpiece, the values of γ for the low, middle and high arc current settings are 0.24, 0.59 and 1 respectively, indicating that the middle arc current setting gives an optimal result. Further comparisons of the temperature distribution and welding profiles for the 5 mm workpiece at the three arc current settings are shown in Fig. 6. As can be seen, when the current increases from 95 to 250 A, the maximum temperature in the arc increases from 14 600 to around 17 000 K, while the weld pool becomes deeper and wider. The current of 95 A generates insufficient heat for significant penetration of the weld, while 250 A generates an overly large molten region that penetrates through the bottom of the workpiece.

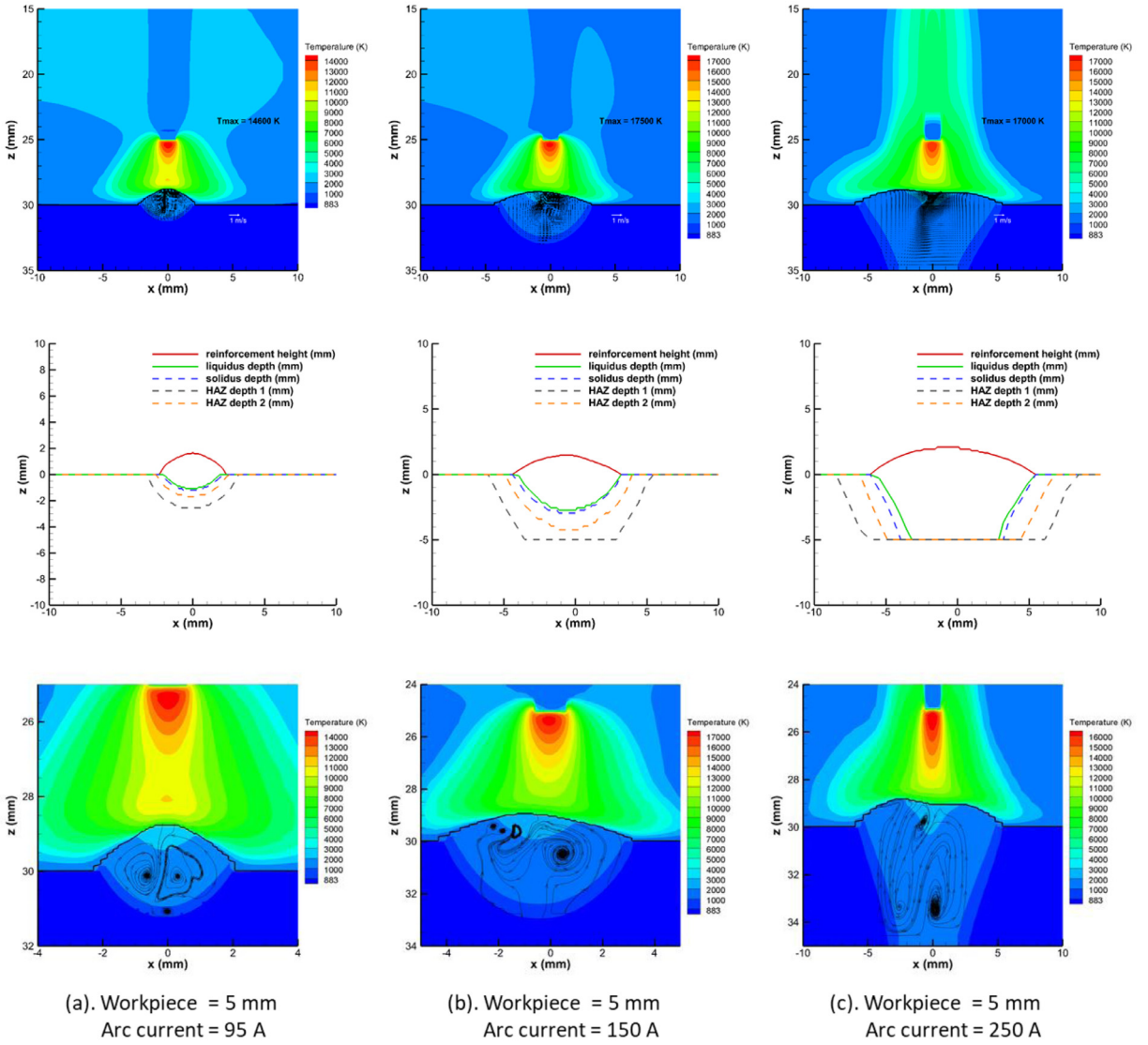


Fig. 6. Comparison of temperature (top row) and weld profile (middle row) for 5 mm workpiece (AA5754) welded at different arc currents for bead-on-plate geometry. The bottom row shows the streamlines of the flow field in the weld pool. The 'HAZ depths' in the weld profile plots correspond to the heat-affected zone reaching temperatures of at least 400 and 500 °C.

For the 8 mm workpiece, the high arc current setting increases the value of the ratio γ to 0.58, which is within the target range, while the lower arc current settings cannot generate a deep enough weld pool for such a thick workpiece.

Fig. 7 shows the predicted optimal welding profiles for each workpiece thickness (3 mm, 5 mm and 8 mm). The simulation results are consistent with the experimental knowledge that welding a thicker workpiece requires a higher arc current to generate higher temperature field and a higher wire feed rate to maintain the arc length. The welding speed can also be reduced to allow a longer penetration time for a larger molten pool.

To summarize the model optimization process presented in this section, a few important points should be emphasized. Firstly, a set of objective indicators is important for MIG welding parameters optimization. Here we used the ratio of the welding depth to the workpiece thickness as an indicator, but this can be extended to include multiple indicators, such as combining this depth ratio with another ratio, that of the reinforcement height to the bead width, and so on, based on the particular application that is being investigated. Secondly, the model-based optimization allows a quantitative comparison of predicted performance for different welding parameters, and hence selection of the optimal settings for these parameters. In this example, the relationship of the arc current settings to the workpiece thickness has been quantified through parameter optimization, providing insights that will help to improve the experimental process. In the next section, another model optimization example will be provided: selection of the optimal arc current settings for different workpiece alloys. This can be very useful for indicating appropriate experimental parameters for welding uncommon or previously unused metals and

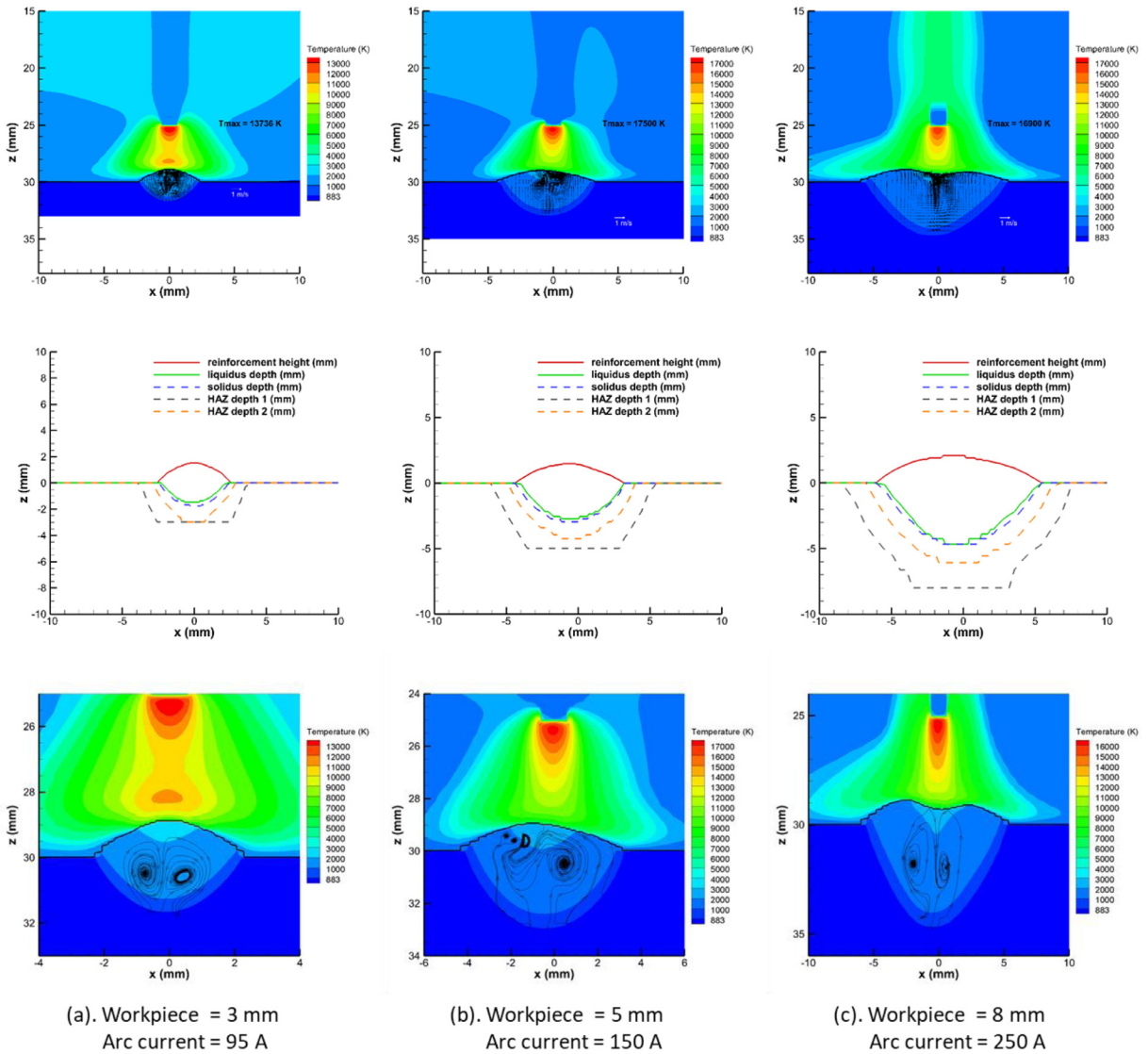


Fig. 7. Comparison of temperature (top row) and weld profiles (middle row) for different workpiece thicknesses simulated at the optimal arc current settings (row in bold text for each thickness shown in Table 4). The bottom row shows the streamlines of the flow field in the weld pool.

alloys. Finally, although we have focused on the arc current, wire feed rate and welding speed, the model optimization approach can be used for multiple parameters. With the aid of high-performance computing, the model-based optimization can be effectively applied for a large set of parameters and complicated objective functions.

4.3. Finding optimal arc current settings for different metal alloys

Alloy selection is another important topic in MIG welding. As illustrated in Fig. 2, the material properties, particularly thermophysical properties, of metal alloys to be welded are key input parameters for the solver. Any change in alloy type of the wire or workpiece will result in changes to the metal properties. It is, therefore, necessary to adjust welding parameters for effective results. This section compares the predicted welding performance among three types of commonly used alloys: aluminium–silicon–magnesium (Al–Si–Mg) [32], aluminium–magnesium–zinc (Al–Mg–Zn) [32] and magnesium alloys [33]. The specific alloy combination of wire electrode and workpiece for each case is given in Table 5.

With the prior knowledge of the Al–Si–Mg alloy presented in 4.1 and 4.2, the aim of this study was to find the optimal arc current settings for Al–Mg–Zn alloys and an Mg alloy. Following the similar optimization process presented in Section 4.2, the numerical simulations were conducted for three different alloys at two different arc currents (95 A and -120 A) and two different welding speeds (0.7 m/min and 0.9 m/min); see Table 6.

Table 5
Alloy selections for wire electrode and workpiece.

Alloy group	Wire electrode	Workpiece
Al–Si–Mg	AA4043 (Al + 5.3 wt% Si)	AA5754 (Al + 3.1 wt% Mg)
Al–Mg–Zn	ER5087 (Al + 4.85 wt% Mg)	7N01 (Al + 1.5 wt% Mg + 4.5 wt% Zn)
Mg-alloy	AZ31 (Mg + 3 wt% Al + 1 wt% Zn)	AZ31 (same as wire)

Table 6
The main welding parameters used for optimization study on different alloys.

Parameter	Value
Arc current (A)	95, 120
Wire feed rate (m/min)	4.3, 5.4
Welding speed (m/min)	0.7, 0.9
Gas flow (L/min)	14.16
Droplet frequency (1/s)	93
Workpiece thickness (mm)	3
Wire anode alloy	(a). AA4043, (b). ER5087, (c). AZ31
Workpiece cathode alloy	(a). AA5754, (b). 7N01, (c). AZ31
Wire electrode angle	90° (vertical)
Weld geometry	Bead-on-plate

Table 7
Model predicted results for different workpiece alloys simulated at different arc currents and welding speeds. The optimal settings are highlighted in bold for each alloy type.

Wire/workpiece alloys	Arc current (A)	Wire feed rate (m/min)	Welding speed (m/min)	RH (mm)	WD (mm)	$\gamma = WD/T$
AA4043/AA5754	95	4.3	0.7	1.8	1.86	0.62
	95	**4.3**	**0.9**	**1.54**	**1.76**	**0.59**
	120	5.4	0.7	1.8	3	1
	120	5.4	0.9	1.37	3	1
ER5087/7N01	95	4.3	0.7	1.8	1.55	0.52
	95	**4.3**	**0.9**	**1.54**	**1.75**	**0.58**
	120	5.4	0.7	1.93	2.46	0.82
	120	5.4	0.9	1.48	3	1
AZ31/AZ31	95	4.3	0.7	1.71	1.11	0.37
	95	4.3	0.9	1.37	1.63	0.54
	120	5.4	0.7	1.93	1.19	0.4
	120	**5.4**	**0.9**	**1.65**	**2.09**	**0.7**

Table 8
The main welding parameters used for the AZ31 experiment on a butt-joint.

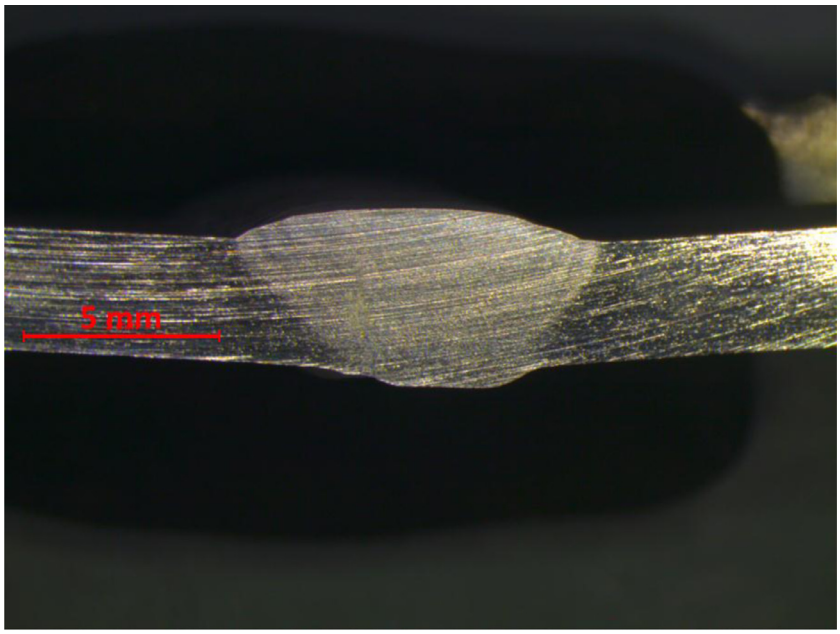
Parameter	Value
Arc current (A)	130
Gas flow (L/min)	14.16
Wire feed rate (m/min)	5.5
Welding speed (m/min)	1
Workpiece thickness (mm)	3
Wire anode alloy	AZ31
Workpiece cathode alloy	AZ31
Wire electrode angle	90° (vertical)
Wire diameter (mm)	1

Table 7 compares the predicted weld depths, with the optimal settings for each alloy shown in bold. For AA4043/AA5754 case, the arc current 120 A is too high for both welding speeds, giving penetration through the bottom of the workpiece with $\gamma = 1$. An arc current of 95 A generates appropriate weld depths for both welding speeds (γ is approximately 0.6 for both speeds), but the lower welding speed (0.7 m/min) generates a slight higher reinforcement height (1.80 mm) compared to 1.54 mm for the higher speed. For the case of this alloy, the welding speed of 0.9 m/min is a better choice since it will give a shorter welding time and lower wire use, thereby reducing cost.

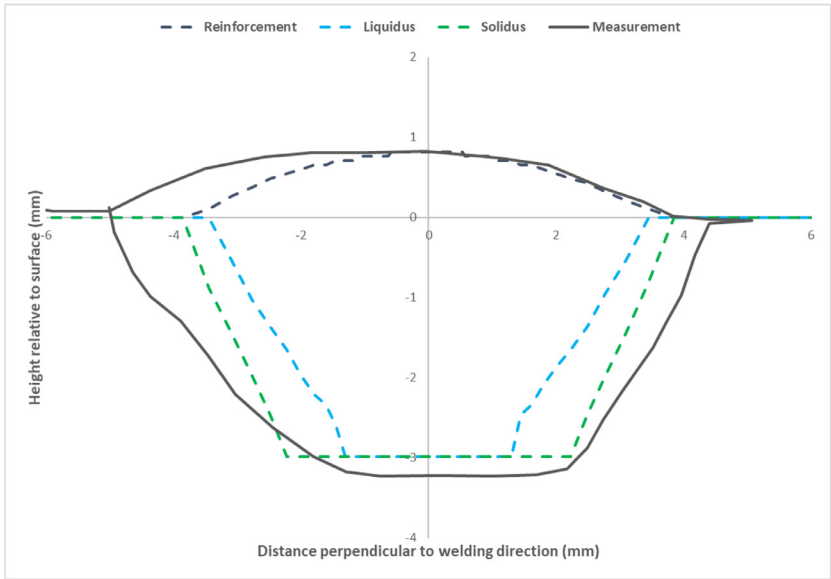
For welding of the Al–Mg–Zn alloy (ER5087/7N01), the simulation results are very similar to Al–Si–Mg case since they are both Al alloy series with similar thermophysical properties. The similar low arc current (95 A with 4.3 m/min wire feed rate) and higher welding speed (0.9 m/min) would be a good selection, achieving nearly 60% penetration of the workpiece.

Table 9
Numerical comparison of the measured weld dimensions with the model predicted values for AZ31 butt joint.

	RH (mm)	BW (mm)	WD (mm)	WD/T	RH/BW
Measured	0.826	9.2	3.2	1.06	0.09
Predicted	0.82	8.2	3	1	0.1



(a). A cross-sectional graph of AZ31 butt joint



(b). weld profiles overlap of measured (solid lines) with the predicted (dash lines)

Fig. 8. Comparison of the experimental measurements and the model predictions for AZ31 bead-on-plate welding.

For the Mg alloy AZ31, the simulation results predict a shallower weld for 95 A arc current for both welding speeds ($\gamma < 0.55$). An arc current of 120 A produces a slight deeper weld pool. The welding speed affects both the reinforcement height and the weld depth. For the selected optimization target, the optimal settings are arc current 120 A and welding at higher speed (0.9 m/min), which give good penetration rate of the AZ31 workpiece (γ close to 0.7).

There are some technical challenges in welding of Mg alloys because of their particular thermophysical properties such as low boiling temperature and high vapour pressure. The boiling point and the latent heat of vaporization of magnesium are much lower than those of aluminium, which lead to higher Mg vapour pressure and higher radiative cooling of the arc. This in turn leads to a cooler arc and lower heat transfer to the workpiece. These property differences affect the weld pool shape; as shown in our simulations, the AZ31 weld is shallower than the Al alloy weld for the same arc current.

Solidification shrinkage of Mg during cooling can also cause significant deformation of the welded parts. Good experimental data becomes important for the model validation on Mg alloy welding. Using available MIG welding experimental data on AZ31 from a rail industry application, we compare the weld cross-section predicted by *ArcWeld* to the experimental measurement. Fig. 8(a) shows an experimental cross-sectional macrograph of a butt-joint weld of AZ31. Deformation can be observed in the weld pool region. The welding parameters used in the experiment are somewhat different from those used in our optimization study and are shown in Table 8.

The numerical simulation for AZ31 has been run using the same parameter values given in Table 8. Comparisons of the measured reinforcement height (RH), weld depth (WD) and bead width (BW) with the predicted values are shown in Table 9. Fig. 8(b) compares the predicted weld pool profiles to the measured profiles. The predicted weld pool shows reasonably good agreement with the experimental result for the 130 A arc current, with full penetration in both cases.

In some applications, full penetration is desirable, particularly for butt welds. However, the greater heat input required can lead to increased residual stress and distortion, which can be observed in Fig. 8(a). Our optimization study on Mg alloy suggests that a lower arc current input of 120 A and welding speed 0.9 m/min are preferable for AZ31. This optimal setting has lower heat input but still gives good penetration ($\gamma = 0.7$).

In summary, this section demonstrated the model-based parameter optimization for dealing with different welding needs, such as for finding the optimal arc current settings for Mg alloy AZ31. The experimental data benchmarking validated the model's capability in the prediction of the different weld pool behaviours between Mg alloy and Al alloy due to the different thermophysical properties. The case study on arc current and workpiece thickness showed that the MIG modelling approach presented is able to perform a quantitative study of the relationship between important MIG welding parameters for process optimization. The last example demonstrated that our models can quickly provide crucial information in exploring new metal types for MIG welding.

5. Conclusion

Parameter optimization based on the existing computational models has been presented in this paper for arc welding process simulation and prediction of thermal and physical properties of the arc and the metal. Using the arc welding modelling software, the important welding process parameters can be optimized to produce a weld pool with the required properties. In addition, the plasma interaction with the wire and workpiece and the dimensions of the mushy zone, which are impractical to measure experimentally, have been incorporated into the modelling software to improve the accuracy of the model's predictions.

Experimental benchmarking and example cases presented in the paper have shown an effective way to identify key parameters in the complicated welding process and quantify the inter-relationship of those main factors, as well as their effects on the welding predictions. Through extended cross-industry collaborations, our MIG welding modelling software has been used by our automotive and rail industrial partners in different applications. The model-based parameter optimization offers a competitive solution for practical improvement of the welding process.

The key contributions of this work can be summarized as follows:

- Unlike general-purpose CFD modelling software, *ArcWeld* is a specialized MIG welding modelling software. As well as including a sophisticated 3D magnetohydrodynamic model to deal with interactions between the arc plasma, wire electrode and workpiece, it has been deployed as a user-friendly software for commercial use.
- The presented model optimization process using *ArcWeld* software can be used either as a quantitative tool for welding pool shape and dimension assessment or as a predictive tool for guiding experimental design and solving reverse engineering problems for customized welding requirements.
- The presented work demonstrates an effective virtual experimental approach for welding process optimization.

Further development of the model will focus on application-specific requirements of welding geometry and metal type and customized prediction tools for different welding needs.

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